Uni[MASK]: Unified Inference in Sequential Decision Problems

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Full paper here 👆

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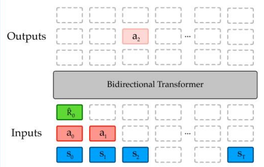
Background: BERT and Randomized Masking

In NLP, having the model *predict masked input words* has been a powerful idea for learning general-purpose representations of language.

Can the same idea be useful in sequential decision problems?

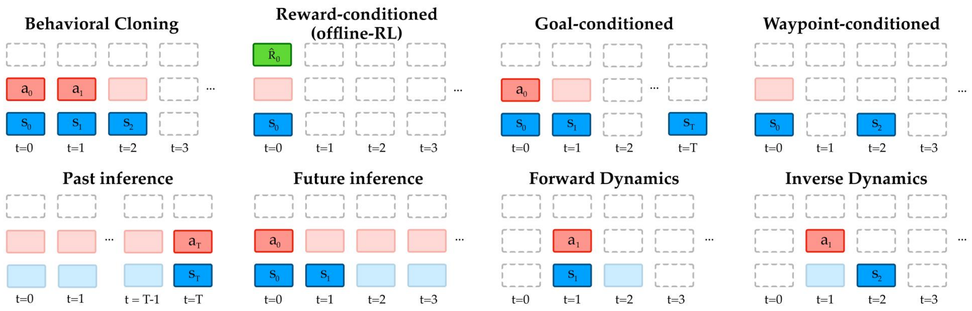


Tokens are states, actions, and rewards. Training for predicting masked inputs amounts to training on different *inference tasks* (e.g. BC, future prediction, etc.).

Our question: *by training on random mask prediction tasks, can we train a single multi-task model for any task in a sequential decision problem?*

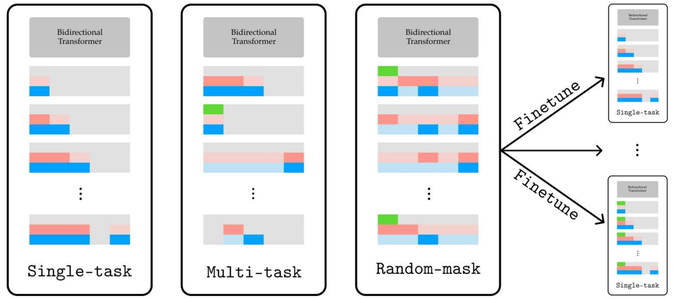
Inference Tasks as Masking Schemes

Key idea: any inference in sequential decision problems *can be expressed as a sequence masking.*
The model receives a subset of observed tokens (state, action, reward), and has to predict some other subset.



Many possible training regimes...

Train ***specialized multi-task***, or ***any-task*** models with a single, unified architecture



1. Can one train on all possible inference tasks and learn to do *anything* sensible?

Training a single model on all tasks (via random masking) not only works, but can do as well or better than specialized models!

2. Which training regime is best if you care about a single downstream inference task?

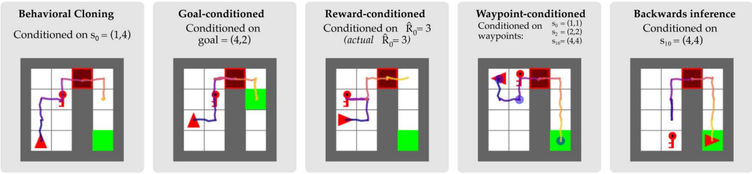
While pre-training on multiple maskings does well, fine-tuning this general model one can improve performance further.

3. Which training regime is best if you care about *n* downstream inference tasks?

Training on all possible tasks (with random masking) can sometimes enable to do better than training on the n tasks of interest.

Results: a single set of model parameters for *all* tasks

Discrete Minigrid Environment



Normalized validation loss (column-wise)

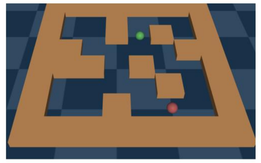
Training task \ Evaluation task	Behavior Cloning	Reward Conditioned	Goal Conditioned	Waypoint Conditioned	Past Inference	Future Inference	Forwards Dynamics	Inverse Dynamics
Behavior Cloning	1	1.037	1.212	1.614	1.602	1.575	1000	5.644
Reward Conditioned	1.04	1.022	1.252	1.674	1.589	1.584	1000	5.61
Goal Conditioned	1.175	1.213	1.104	1.534	1.653	1.687	1000	5.968
Waypoint Conditioned	1.259	1.301	1.167	1.249	1.682	1.75	1000	6.059
Past Inference	1.423	1.469	1.645	2.126	1.061	1.681	1000	3.633
Future Inference	1.025	1.056	1.23	1.652	1.81	1.045	1000	4.652
Forwards Dynamics	1.433	1.484	1.7	2.22	4.208	3.97	2.616	2.48
Inverse Dynamics	1.401	1.439	1.684	2.197	2.629	2.141	1000	1.05
Multi-task (All the above)	1.017	1.034	1.124	1.434	1.102	1.072	1000	1.482
Random-mask	1.09	1.045	1.039	1.044	1.005	1.013	539	1.151
Random M. +Finetune	1.03	1	1	1	1	1	1	1

① random masking pre-training + finetuning > ③ Specialized models

② Random masking pre-training (zero-shot) ≈ ③ Specialized models

A single Uni[MASK] model can be used for arbitrary inference tasks out-of-the-box when trained on random masking.

Continuous Control
(Mujoco Maze2D)



Model	Context Length 5 BC	Context Length 5 RC	Context Length 10 BC	Context Length 10 RC
Uni[MASK] Models				
Uni[MASK]- `single-task`	2.66 ± 0.03	2.64 ± 0.02	2.47 ± 0.04	2.41 ± 0.05
Uni[MASK]- `multi-task` (BC & RC)	2.65 ± 0.01†	2.68 ± 0.01†	2.39 ± 0.03†	2.39 ± 0.03†
Uni[MASK]- `multi-task` + finetune	2.73 ± 0.01	2.74 ± 0.01	2.42 ± 0.04	2.42 ± 0.03
Uni[MASK]- `random-mask`	2.19 ± 0.09†	2.20 ± 0.09†	2.29 ± 0.07†	2.31 ± 0.06†
Uni[MASK]- `finetune`	2.67 ± 0.03	2.73 ± 0.01	2.55 ± 0.03	2.61 ± 0.03
Other architectures				
Feedforward Neural Network	1.68 ± 0.07	1.53 ± 0.08	1.83 ± 0.06	1.88 ± 0.06
Decision Transformer [7]	1.13 ± 0.07	1.49 ± 0.04	1.58 ± 0.06	1.70 ± 0.07
Our Decision-GPT model	2.66 ± 0.01	2.32 ± 0.05	2.74 ± 0.01	2.73 ± 0.02

Pre-training and fine-tuning on single-tasks achieves the best perfomance among Uni[MASK] regimes.

Our unidirectional single-task variant achieves best performance overall.

Limitations and Future Work

- Scale up to more complex environments
- Systematically investigate how multi-task performance varies with larger context windows
- Train models to roll out trajectories conditioned on other useful properties (e.g., style-matching).

Conclusions:

The Uni[MASK] framework enables models to:

1. Naturally represent any inference task
2. Easily perform multi-task training with no architecture changes
3. Train single models to do *all tasks* with random-mask pre-training, which perform similarly to specialized models and often surpass them after fine-tuning.